

High-redshift JWST massive galaxies and the initial clustering of supermassive primordial black holes

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In this paper, we show that the initial clustering of supermassive primordial black holes (SMPBHs) **beyond a Poisson distribution can efficiently enhance the matter power spectrum, and thus the halo mass function.** As a result, the population of initially clustered SMPBHs with $M_{\text{PBH}} \sim 10^9 M_\odot$ and the fraction of energy density $f_{\text{PBH}} \sim 10^{-3}$ (consistent with current constraints on SMPBHs) has the potential to naturally explain high-redshift massive galaxies observed by the James Webb Space Telescope.

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I. INTRODUCTION

Recent observations with the James Webb Space Telescope (JWST) have discovered a large number of quasars powered by supermassive black holes (SMBHs) already in place within the first few hundred million years after the big bang [1–14]. The existence of such SMBHs at very high redshifts poses a significant challenge. It is commonly assumed that SMBH seeds initially form, such as seed black holes expected from Pop III stellar remnants [15], and then grow into SMBHs over time until the redshifts of observations. However, even if they accrete at the maximal Eddington rate, there does not seem to be enough cosmic time to grow them sufficiently.¹ In this scenario, the direct birth of SMBHs in the very early Universe, i.e., SMPBHs, becomes rather appealing.

The initial clustering of stellar-mass PBHs has been ruled out by microlensing and Lyman- α forest observations [17]. However, it seems that SMPBHs prefer an initially clustered spatial distribution. In the standard mechanism that PBHs are from a direct collapse of large amplitude perturbations after horizon entry [18–22], non-Gaussian primordial perturbations seems to necessary to evade the μ -distortion constraints for SMPBHs [23–30], while the non-Gaussianity of primordial perturbation will possibly lead to

the initial clustering of SMPBHs [31–35]. In the mechanism [36,37] where SMPBHs are sourced by the inflationary bubbles and have a multipeak mass spectrum, the PBHs can be spatial-clustered naturally in multistream inflation [38,39], see also [37,40,41]. The PBHs from Affleck-Dine mechanism also show a strong clustering [42,43], see also [44].

It has been showed in [36,37,45–47] that the merger of SMPBHs might be the source of nano-Hertz gravitational wave background recently detected by pulsar timing arrays (PTA) collaborations [48–51], however, the distribution of SMPBHs must exhibit some clustering to magnify the merger rate of SMPBH binaries. On the other hand, considering the nonzero angular auto-correlation function of quasars [52,53], models predicting the Poisson distribution of SMPBHs are ruled out. The Sloan Digital Sky Survey [54] and the 2dF QSO redshift survey [55] have measured the auto-correlation function of quasars up to $z \approx 4$ [56,57]. The autocorrelation function of faint quasars at $z \approx 6$ was also recently estimated, benefiting from the high sensitivity of the Subaru High- z Exploration of Low-Luminosity Quasars survey [58].

Recently, based on 14 galaxy candidates with masses in the range 10^9 – $10^{11} M_\odot$ at $7 < z < 11$ identified in the JWST Cosmic Evolution Early Release Science program, [59] found that the cumulative stellar mass density at redshift $z^{\text{obs}} = 8$ for the stellar mass density $M_* \geq 10^{10} M_\odot$ is

$$\rho^{\text{obs}}(M_* \geq 10^{10} M_\odot) \simeq 1.3 \times 10^6 M_\odot \text{Mpc}^{-3}. \quad (1)$$

This result is hardly reconcilable with the standard Λ cold dark matter model (Λ CDM) expectation, which would require an implausible high star formation efficiency, even

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¹It might be thought the hypothesis that heavy seeds could be direct collapse black holes (DCBHs) might alleviate this problem. However, DCBHs have long been considered too rare to account for the entire observed SMBH population, see also [16] for recent numerical simulations.

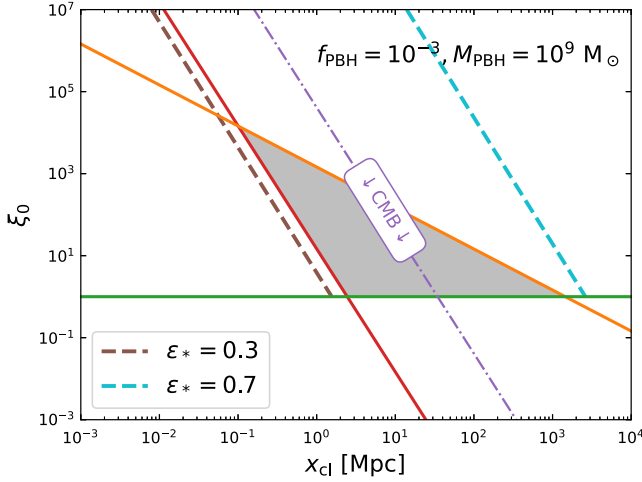


FIG. 1. The parameter space of (ξ_0, x_{cl}) for initially clustered SMPBHs explaining the JWST observation, where we have assumed $M_{\text{PBH}} = 10^9 M_\odot$ and $f_{\text{PBH}} = 10^{-3}$. The gray shaded region represents the parameter space allowed by the model itself, constrained by Eqs. (6) (green), (7) (red), and (8) (orange), respectively. The purple line represents the observation constraint from the CMB, Eq. (22). The brown (cyan) line indicate clustering SMPBH which can explain the comoving cumulative stellar mass density for the star formation rate $\epsilon_* = 0.3$ (0.7). We do not consider the enhancement of the PBH isocurvature power spectrum at $k > k_{\text{cut,cluster}}$ [see Eq. (19) and the discussion nearby], and any correlation between the adiabatic and isocurvature modes is ignored here. Therefore, the estimates of the required star formation efficiency shown here are the most conservative results in clustered SMPBH model.

larger than the cosmic baryon mass budget in collapsed structures. This conflict has inspired lots of relevant studies [60–72].

However, if SMPBHs do exist, they would enhance the halo mass function and thus boost the early formation of massive galaxies [61, 73–76]. In this paper, we explore the effect of the initial clustering of SMPBHs. It is found that the initial clustering can further enhance the halo mass function at the high mass tail. As a result, the population of initially clustered SMPBHs with $M_{\text{PBH}} \sim 10^9 M_\odot$ and $f_{\text{PBH}} \sim 10^{-3}$ (consistent with current constraints on SMPBHs, e.g., [77]) has the potential to naturally explain high-redshift massive galaxies observed by JWST. The results are summarized in Fig. 1.

II. MODELING INITIAL CLUSTERING OF PBHS

The clustering of PBHs beyond the Poisson distribution is described by its correlation functions, and the precise shape is model-dependent. Here, for simplicity the correlation function of PBHs is modeled as [17]

$$\xi_{\text{PBH}}(x) = \begin{cases} \xi_0 & \text{if } x \leq x_{\text{cl}}, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where x_{cl} is the comoving size of the gravitationally bound cluster. The corresponding PBH clusters follow a Poisson distribution at much large scales.

It has been argued in [78] that PBHs are anticorrelated at short distances as one should expect that there is at most one PBH per horizon volume, i.e., $\xi_{\text{PBH}}(x) \simeq -1$ for $x \lesssim x_{\text{exc}}$, where the comoving size x_{exc} of the small-scale exclusion volume is approximately the comoving Hubble radius at formation time of PBHs,

$$x_{\text{exc}} \simeq x_H \simeq \left(\frac{M_{\text{PBH}}}{M_\odot} \right)^{1/2} \frac{1}{4 \times 10^6 \text{ Mpc}^{-1}}. \quad (3)$$

We have verified that the spatial exclusion condition has negligible effect on our subsequent calculations ($x_{\text{exc}} \ll x_{\text{cl}}$), allowing us to safely adopt Eq. (2).

Here, for analytical viability, we adopt a monochromatic PBH mass function with a mass parameter M_{PBH} and the fraction of dark matter (DM) $f_{\text{PBH}} = \Omega_{\text{PBH}}/\Omega_{\text{DM}}$. The average energy density of PBHs is $\bar{\rho}_{\text{PBH}} = \bar{n}_{\text{PBH}} M_{\text{PBH}}$, where the average (comoving) number density is

$$\bar{n}_{\text{PBH}} \simeq 3.3 \times 10^{10} f_{\text{PBH}} \left(\frac{M_{\text{PBH}}}{M_\odot} \right)^{-1} \text{ Mpc}^{-3}. \quad (4)$$

In such a PBH cluster, the number of PBHs is

$$N_{\text{cl}} = 1 + \bar{n}_{\text{PBH}} \int d^3x \xi_{\text{PBH}}(x), \quad (5)$$

see [17] for details, which is approximately $N_{\text{cl}} \simeq \frac{4\pi}{3} \bar{n}_{\text{PBH}} \xi_0 x_{\text{cl}}^3$. Thus we have that

$$\xi_0 \simeq \frac{N_{\text{cl}} / \frac{4\pi}{3} \pi x_{\text{cl}}^3}{\bar{n}_{\text{PBH}}} \equiv \frac{\bar{n}_{\text{PBH,cl}}}{\bar{n}_{\text{PBH}}} > 1 \quad (6)$$

represents the ratio of the number density of PBHs within the cluster to the average number density approximately. The Poisson distribution corresponds to $\xi_{\text{PBH}}(x) = 0$, or equivalently, $N_{\text{cl}} = 1$.

In clustered scenarios, we should assume that $N_{\text{cl}} \gtrsim 3$, i.e.,

$$\xi_0 \gtrsim 1.4 \times 10^{-11} f_{\text{PBH}}^{-1} \left(\frac{M_{\text{PBH}}}{M_\odot} \right) \left(\frac{x_{\text{cl}}}{\text{Mpc}} \right)^{-3}. \quad (7)$$

Additionally, it is required that the cluster does not collapse into a heavy PBH with mass of $N_{\text{cl}} M_{\text{PBH}}$, which suggests [17, 79]

$$\xi_0 \lesssim 60 f_{\text{PBH}}^{-1/3} \left(\frac{C}{200} \right)^{-1/6} \left(\frac{x_{\text{cl}}}{\text{Mpc}} \right)^{-1} \quad (8)$$

according to the hoop condition $r_{\text{cl}} \lesssim 2GM_{\text{cl}}$, where $C = \mathcal{O}(1-100)$ is a constant of proportionality, r_{cl} and M_{cl} are the physical radius and mass of the cluster.

III. THOSE ENHANCED BY CLUSTERING SMPBHs

A. Power spectrum

The density fluctuation of PBHs is

$$\delta_{\text{PBH}}(\mathbf{x}) \equiv \frac{\delta_{\text{PBH}}(\mathbf{x})}{\bar{\rho}_{\text{PBH}}} = \frac{1}{\bar{n}_{\text{PBH}}} \sum_i \delta_D(\mathbf{x} - \mathbf{x}_i) - 1, \quad (9)$$

noting that these PBHs are spatially discrete objects. $\delta_D(\mathbf{x})$ is the three-dimensional Dirac distribution. The corresponding two-point correlation function of fluctuation is given by

$$\langle \delta_{\text{PBH}}(0) \delta_{\text{PBH}}(\mathbf{x}) \rangle = \frac{1}{\bar{n}_{\text{PBH}}} \delta_D(\mathbf{x}) + \xi_{\text{PBH}}(x), \quad (10)$$

where $x = |\mathbf{x}|$ is the distance between two PBHs.

The power spectrum of the density perturbations of PBHs is

$$\begin{aligned} P_{\text{PBH}}(k) &= \int d^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}} \langle \delta_{\text{PBH}}(0) \delta_{\text{PBH}}(\mathbf{x}) \rangle \\ &= P_{\text{Poisson}}(k) + P_{\xi}(k), \end{aligned} \quad (11)$$

where $P_{\text{Poisson}}(k) = 1/\bar{n}_{\text{PBH}}$, which is independent of k , comes from the Poisson fluctuation induced by the fluctuation of the number of PBHs, and

$$P_{\xi}(k) = \int d^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}} \xi_{\text{PBH}}(x) = 4\pi \int dx x^2 \frac{\sin kx}{kx} \xi_{\text{PBH}}(x) \quad (12)$$

comes from the clustered distribution of PBHs. Thus considering Eq. (2), we have

$$P_{\xi}(k) = \frac{4\pi \xi_0 [\sin(kx_{\text{cl}}) - kx_{\text{cl}} \cos(kx_{\text{cl}})]}{k^3}. \quad (13)$$

We observe that

$$\lim_{k \rightarrow 0} P_{\xi}(k) = \frac{4\pi}{3} x_{\text{cl}}^3 \xi_0 \simeq \frac{N_{\text{cl}}}{\bar{n}_{\text{PBH}}} > \frac{1}{\bar{n}_{\text{PBH}}} = P_{\text{Poisson}}, \quad (14)$$

which indicates that the contribution of clustering for the power spectrum $P_{\xi}(k)$ exceeds that of the Poisson fluctuation on large scales.

The PBHs contribute to the isocurvature perturbations through $f_{\text{PBH}}^2 P_{\text{PBH}}(k)$. These isocurvature fluctuations then grow during the matter-dominated era, and at $z = 0$, they are given by

$$P_{\text{iso}}(k, z = 0) = \begin{cases} [f_{\text{PBH}} D_{\text{PBH}}(0)]^2 P_{\text{PBH}}(k), & \text{if } k \leq k_{\text{cut}}, \\ 0, & \text{otherwise,} \end{cases} \quad (15)$$

where $D_{\text{PBH}}(z) \simeq (1 + \frac{3\gamma}{2\alpha_-} \frac{1+z_{\text{eq}}}{1+z})^{\alpha_-}$ with $\gamma = (\Omega_m - \Omega_b)/\Omega_m$ and $\alpha_- = (\sqrt{1 + 24\gamma} - 1)/4$ is the growth factor of isocurvature perturbations until today [45,80]. These isocurvature fluctuations are restricted to large scales, because we expect the linear theory to break down at small scales. For initially Poisson-distributed PBHs, the cutoff scale of the spectrum is of order the inverse mean separation between PBHs²

$$k_{\text{cut,Poisson}} = (2\pi^2 \bar{n}_{\text{PBH}})^{1/3}. \quad (16)$$

At scales below $k_{\text{cut,Poisson}}$, we will have a single PBH within the corresponding comoving sphere, signaling the onset of the seed effect prevailing over the Poisson effect [62,80,81]. In the presence of initially clustering, we should replace it with the inverse average separation of clusters³

$$k_{\text{cut,cluster}} = (2\pi^2 \bar{n}_{\text{cl}})^{1/3}, \quad (18)$$

where $\bar{n}_{\text{cl}} = \bar{n}_{\text{PBH}}/N_{\text{cl}}$ is the cluster number density. It is obvious that it corresponds to Eq. (16) when $N_{\text{cl}} = 1$ is satisfied, indicating the self-consistency of our model. Rewriting it with the model parameters (x_{cl}, ξ_0) , we have

$$k_{\text{cut,cluster}} \simeq \left(\frac{3\pi}{2\xi_0} \right)^{1/3} \frac{1}{x_{\text{cl}}}. \quad (19)$$

The Planck results [82] set the upper bound on the primordial isocurvature perturbations on the CMB scale as

$$\beta_{\text{iso}}(k_{\text{CMB}}) \equiv \frac{P_{\text{iso}}(k_{\text{CMB}})}{P_{\text{iso}}(k_{\text{CMB}}) + P_{\mathcal{R}}(k_{\text{CMB}})}, \quad (20)$$

where $P_{\mathcal{R}}(k) = (2\pi^2/k^3) \mathcal{P}_{\mathcal{R}}$ with the power spectrum of curvature perturbations $\mathcal{P}_{\mathcal{R}}$. We conservatively use

²Sometimes, a different choice of cutoff scale $k_{\text{NL}} \simeq (\bar{n}_{\text{PBH}}/f_{\text{PBH}})^{1/3}$ is adopted [45,73], below which the nonlinear effects start to dominate. However, for the SMPBHs with $f_{\text{PBH}} \lesssim 10^{-3}$, our choice, Eq. (16), is more conservative. Additionally, the focus of our investigation is on the effect of the cluster distribution of PBHs compared to the Poisson distribution, so in the comparison we only need to maintain the same cutoff criteria.

³We observe that

$$\bar{n}_{\text{cl}} = \frac{\bar{n}_{\text{PBH}}}{N_{\text{cl}}} < \bar{n}_{\text{PBH}} \xi_0 \simeq \bar{n}_{\text{PBH,cl}}, \quad (17)$$

ensuring that $\bar{n}_{\text{cl}}^{1/3} < \bar{n}_{\text{PBH,cl}}^{1/3}$, implying that the average separation between clusters (the minimal distance between PBHs located in different clusters) is bigger than that between individual PBHs within a cluster.

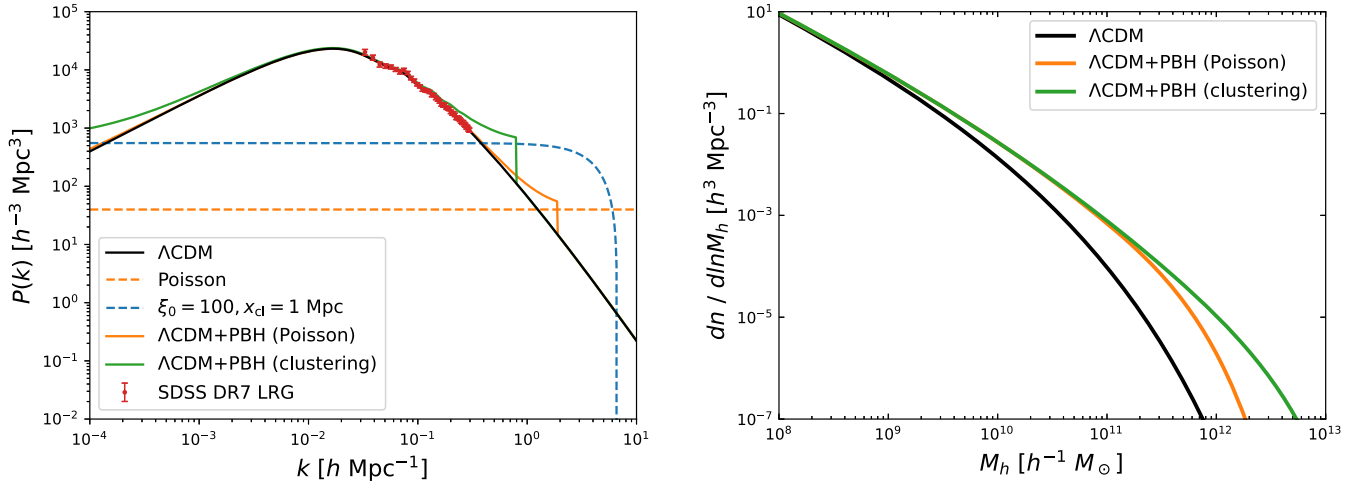


FIG. 2. Left panel: the linear matter power spectrum at $z = 0$ is plotted for Λ CDM (black solid), for Λ CDM with initially Poisson-distributed SMPBHs [$\xi_{\text{PBH}}(x) = 0$, orange solid] and initially clustering SMPBHs (green solid), respectively, where $M_{\text{PBH}} = 10^9 M_\odot$ and $f_{\text{PBH}} = 10^{-3}$. The cutoff Eqs. (16) and (19) are considered respectively. We also plot the isocurvature fluctuation sourced by the Poisson fluctuation (orange dashed) and clustered distribution (blue dashed). The red points with errorbars correspond to observations of Luminous Red Galaxies by the SDSS collaboration [88]. Right panel: the corresponding halo mass function at redshift $z = 8$.

the constraint in [42], i.e., $\beta_{\text{iso}} < 0.035$ at $k_{\text{CMB}} = 2 \times 10^{-3} \text{ Mpc}^{-1}$. On such a large scale, the isocurvature perturbation Eq. (15) can be approximated as

$$P_{\text{iso}}(k) \simeq f_{\text{PBH}}^2 \times \frac{4\pi}{3} x_{\text{cl}}^3 \xi_0, \quad (21)$$

which suggests

$$\xi_0 < 4.1 \times 10^{-2} f_{\text{PBH}}^{-2} \left(\frac{x_{\text{cl}}}{\text{Mpc}} \right)^{-3}. \quad (22)$$

B. Halo mass function according to the power spectrum

We utilize the Sheth-Tormen [83,84] modification to the Press-Schechter formalism [85] with a top-hat window function to calculate the halo mass function. The variance of the smoothed density is

$$\begin{aligned} \sigma^2(M) &= \int P_{\text{tot}}(k, z=0) W^2(kR) \frac{k^2 dk}{2\pi^2} \\ &= \int [P_{\Lambda\text{CDM}}(k, z=0) + P_{\text{iso}}(k, z=0)] \\ &\quad \times W^2(kR) \frac{k^2 dk}{2\pi^2}, \end{aligned} \quad (23)$$

where $W(kR) = \frac{3[\sin(kR) - kR \cos(kR)]}{(kR)^3}$ is the window function, $R = (\frac{3M}{4\pi\rho_m})^{1/3}$ is the comoving radius associated with the mass window M , and $P_{\Lambda\text{CDM}}(k)$ is the power spectrum of adiabatic perturbations for the Λ CDM model without SMPBHs.

The peak significance is defined as

$$\nu_c(M) = \frac{\delta_c}{\sigma(M, z)} = \frac{\delta_c}{D(z)\sigma(M)}, \quad (24)$$

where $D(z) \propto H(z) \int_z^\infty \frac{1+z'}{[H(z')]^3} dz'$ is the linear growth function, normalized so that $D(0) = 1$. Here, we adopt $D(z) \approx 1.27/(1+z)$, which provides a good approximation at $z \gtrsim 1$ [86]. The comoving mean number density of halos $n_h(M)$ per logarithmic mass interval is

$$\frac{dn_h}{d \ln M_h} = \frac{\rho_m}{M_h} \nu_c(M_h) f(\nu_c(M_h)) \frac{d \ln \nu_c(M_h)}{d \ln M_h}, \quad (25)$$

where

$$\nu_c f(\nu_c) = \sqrt{\frac{2}{\pi}} A(p) \left[1 + \frac{1}{(q\nu_c^2)^p} \right] \sqrt{q} \nu_c e^{-q\nu_c^2/2} \quad (26)$$

with $A(p) = [1 + \pi^{-1/2} 2^{-p} \Gamma(0.5 - p)]^{-1}$. Here, we adopt $(q, p) = (0.85, 0.3)$, which has been found to provide an accurate fit to the N -body simulations at high redshifts [87].

In Fig. 2, we depict the power spectrum and the corresponding halo mass function, taking $\xi_0 = 100$ and $x_{\text{cl}} = 1 \text{ Mpc}$ as an example. The effect of SMPBHs is creating a small bulge in the matter power spectrum, with the position and height of the bulge depending on their number density, while for initial clustering SMPBHs, the bulge shifts to the left and its height slightly rises. As a result, the halo mass function is enhanced for $M_h > 10^{11} h^{-1} M_\odot$. The constraints on relevant scales coming from galaxy surveys are showed in red points with error bars. We note that the deviations of our scenario from Λ CDM are at least

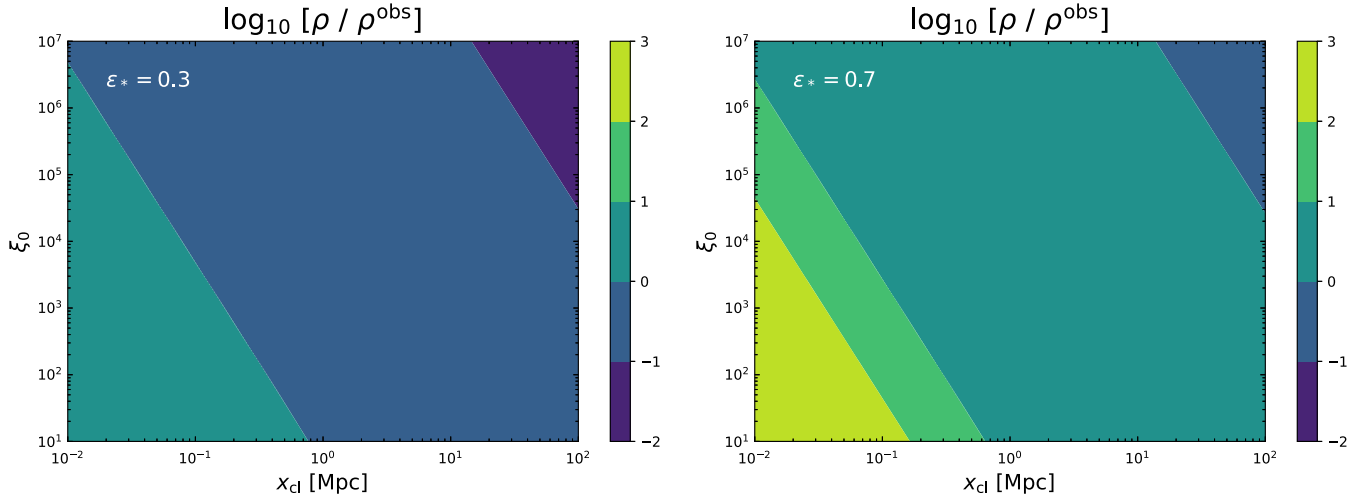


FIG. 3. Left panel: the comoving cumulative stellar mass density at redshift $z^{\text{obs}} = 8$ as a function of (ξ_0, x_{cl}) , assuming a star formation efficiency $\epsilon_* = 0.3$. Right panel: same as the left panel with a different star formation efficiency $\epsilon_* = 0.7$.

compatible with experimental errors, and thus not ruled out by current constraints. In addition, more appropriate selection of model parameters ($\xi_0, x_{\text{cl}}, f_{\text{PBH}}$ and M_{PBH}) can further alleviate the constraints.

IV. EXPLANATION OF JWST OBSERVATIONS

In the following, we explore whether initially clustering SMPBH can explain the massive galaxy candidates by comparing the number density of massive galaxies predicted by our SMPBH model with recent JWST result Eq. (1).

The expected number density of galaxies with $M_* > M_*^{\text{obs}}$, where M_*^{obs} represents the observational threshold of stellar mass, is given by [89]

$$n_{\text{gal}}(M_* \geq M_*^{\text{obs}}) = \int_{M_h^{\text{cut}}}^{\infty} \frac{dn(z^{\text{obs}}, M_h)}{dM_h} dM_h, \quad (27)$$

where $M_h^{\text{cut}} = M_h(M_*^{\text{obs}})$. Here, we speculate that each DM halo contains a single central galaxy, thus the relation between the halo and total stellar mass is $M_h(M_*) = M_*/(f_b \epsilon_*)$, where $f_b = \Omega_b/(\Omega_{\text{DM}} + \Omega_b) = 0.157$ is the baryon fraction and ϵ_* is the star formation efficiency. The comoving cumulative stellar mass density is

$$\rho(M_* \geq M_*^{\text{obs}}) = f_b \epsilon_* \int_{M_h^{\text{cut}}}^{\infty} M_h \frac{dn(z^{\text{obs}}, M_h)}{dM_h} dM_h. \quad (28)$$

The comoving cumulative stellar mass density ρ/ρ^{obs} with respect to (x_{cl}, ξ_0) is presented in Fig. 3, where ρ^{obs} is recent JWST observation Eq. (1). It can be seen that a population of clustering SMPBHs has a larger potential to reproduce the observation results. In contrast, we have $\log_{10}(\rho/\rho^{\text{obs}}) \simeq -1.41$ for ΛCDM model only and $\log_{10}(\rho/\rho^{\text{obs}}) \simeq -0.65$ for

the same SMPBHs but with a Poisson distribution, i.e. the $\Lambda\text{CDM} + \text{PBH}$ (Poisson) model, in the case of $\epsilon_* = 0.3$, both are not sufficient to explain JWST observation, while for a higher $\epsilon_* = 0.7$, we also only have $\log_{10}(\rho/\rho^{\text{obs}}) \simeq -0.34$ and $\log_{10}(\rho/\rho^{\text{obs}}) \simeq 0.24$, respectively. Here, for the $\Lambda\text{CDM} + \text{PBH}$ (clustering) model, we can naturally have $\log_{10}(\rho/\rho^{\text{obs}}) \sim 0$ for $\xi_0 = 10$, $x_{\text{cl}} \sim 1$ Mpc and $\epsilon_* \sim 0.3$.

In particular, there exists a parameter space that renders the model compatible with all constraints on clustering SMPBHs, see the gray shaded region and the purple line in Fig. 1. Taking into account these constraints, we observe that the enhancement of the comoving cumulative stellar mass density caused by the ΛCDM with clustering PBHs is about 2 times larger than that predicted by Poisson-distributed PBHs. This in certain sense reflects the advantage of initial clustering SMPBHs in explaining the high-redshift JWST massive galaxies.

V. DISCUSSION

High-redshift massive galaxies observed recently by JWST can be hardly reconcilable with the standard ΛCDM expectations. The existence of SMPBHs can enhance the matter power spectrum and thus the halo mass function, which helps to alleviate this conflict.

However, SMPBHs can cluster, see [37], and the merger of SMPBHs initially clustered might be the source of nano-Hertz gravitational wave background recently detected by PTA collaboration. This clustering not only sheds light on the origins of SMBHs but also accelerates the evolution of structures in the early universe. In this paper, we calculated the effect of the initial clustering SMPBHs on the matter power spectrum and the halo mass function, and found that compared with the Poisson-distributed SMPBHs model, the population of clustering SMPBHs with $M_{\text{PBH}} \sim 10^9 M_{\odot}$

and $f_{\text{PBH}} \sim 10^{-3}$ can significantly accelerate the early formation of massive galaxies, and thus naturally explain recent JWST observation with a lower star formation efficiency. The collective results of our analysis is showed in Fig. 1.

The exploration of SMPBHs as potential progenitors of SMBHs that power high-redshift quasars presents intriguing insights into the evolution of our Universe. In this case the concept of initially clustered SMPBHs might be closely related to not only the evolution of our primordial Universe but also the spatial distribution of

massive objects and the birth of cosmic structures. The relevant issues are worth of further investigation.

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